7 April 26

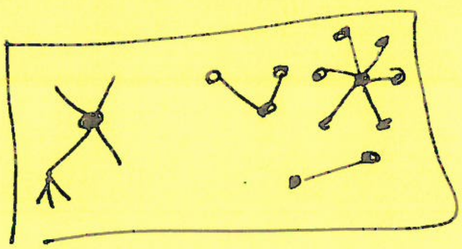
<u>Reminders</u>:

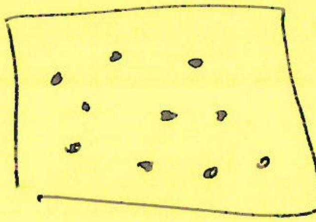
Tree (in graph theory): a graph that has no cycles (aka $\nexists$ a nontrivial path that starts/ends at the same place)

ex: binary search tree: root + 2 children per node (unless a leaf node)

parent/child relationship

Forest: a set of trees

Ex: 

Another Ex: 

EX3 

a tree

<u>Minimum</u> Spanning <u>Tree</u>:

Given a graph $G=(V,E,\omega)$ a MST for G is a tree $T=(V_T,E_T)$ such that

(i) $V_T = V$ (spans all the vertices)

(ii) of all spanning trees, this has minimum cost

$$c(T) = \sum_{e \in E_T} \omega(e)$$

<u>Connected components</u>: can partition V into CC's

- v, w are in the same connected comp if $\exists$ path from v to w.

Question 2:

What is "the issue":

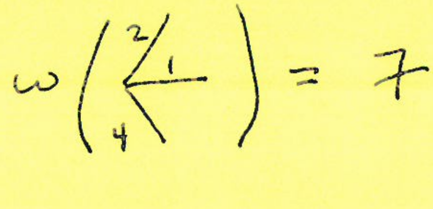
or

} How to pick among "the MSTs" so we're always finding the same one.

We can't

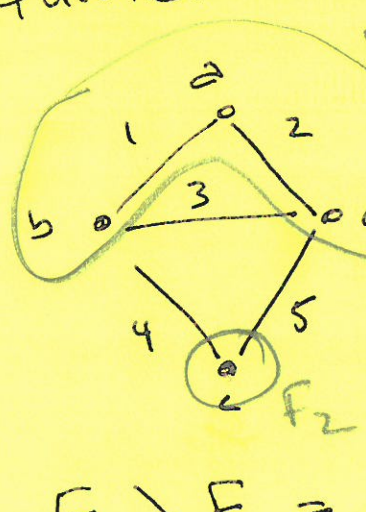
There is no cannonical MST.

So, the assumption that edge weights are unique will work.



Question 3:

$F_1 = \{a, b, d, ab, ad\}$



$F_2 = \{c\}$

$E \setminus F = \{\cancel{ab}, \cancel{ad}, bd, bc, cd\} \setminus \{\cancel{ab}, \cancel{ad}\}$

↑ edge set ↑ graph

$= \{bd, \underbrace{bc, cd}_{\text{different conn. components.}}\}$

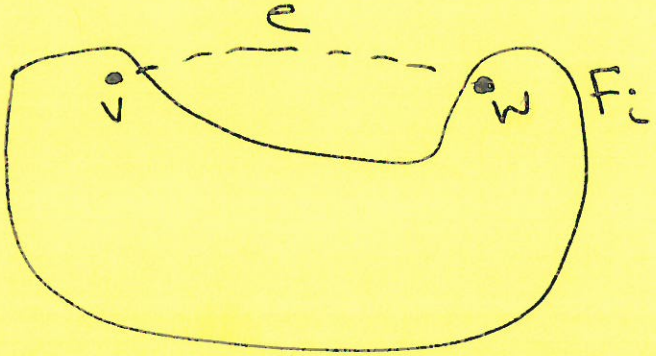
↑ ✓

$e = \{bd\}$ meets criteria → but not in MST.

Question 3

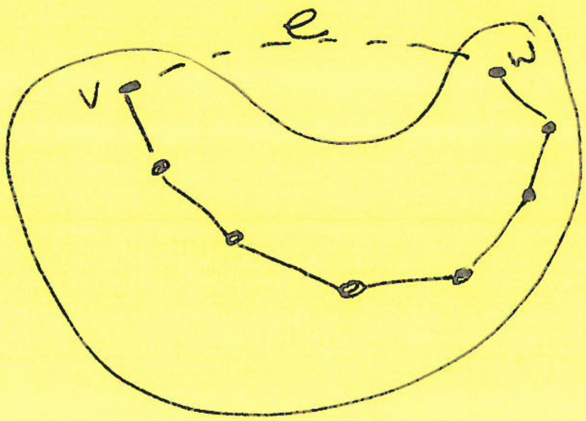
Why are useless edges useless?

Consider F_i the connected comp containing the endpts of $e = (v, w)$



Because v, w are in same conn comp

$\Rightarrow \exists$ a path from v to w in F_i.



(length > 1 b/c otherwise, we'd have a double edge)

$F_i \cup \{e\}$ contains a loop $\Rightarrow$ not a subgraph of MST

Know all edges of F_i are in MST

$\Rightarrow e \notin$ MST.